PRASHANT TEWARI

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EDUCATION

***Leon H. Charney School of Marine Sciences, University of Haifa**, Haifa, Israel.* 2023-prusuing
PhD in Marine Biology

***Leon H. Charney School of Marine Sciences, University of Haifa**, Haifa, Israel.* 2021- 2023
Masters in Marine Biology, Dean of Graduates Studies Honors
Percentage: 94

***Sam Higginbottom University of Agriculture, Technology & Sciences**, Prayagraj, India* 2016-2020
Bachelor of Science (Honours) Biotechnology
C.G.P.A - 7.78

RESEARCH INTEREST

Development Biology, Biomineralization, Genetics, Cell and Molecular Biology

RESEARCH EXPERIENCE

Regulation of Evolution and Development laboratory, University of Haifa Israel. 2023 – Pursing
PhD student
Supervisor: Prof. Smadar Ben Tabou de Leon, Associate Professor, Department of Marine Biology.
Thesis: Deciphering the role of cytoskeleton remodelling and vesicular transport in sea urchin larval biomineralization.

- My PhD research focuses on understanding the cellular and molecular mechanisms underlying skeletogenesis in sea urchin embryos, with particular emphasis on actomyosin regulation, vesicle trafficking, and biomineralization dynamics.
- I use live-cell imaging, quantitative image analysis, and pharmacological perturbations, to elucidate how cytoskeletal and signalling networks coordinate mineral deposition during embryonic development.

Regulation of Evolution and Development laboratory, University of Haifa Israel. 2021- 2023
Masters Student
Supervisor: Prof. Smadar Ben Tabou de Leon, Associate Professor, Department of Marine Biology.
Thesis: p21 Activated Kinases Expression, Regulation, and their Roles in Sea Urchin Development and Skeletogenesis

- I worked on characterising the role of p21-activated kinase (Paks) genes in sea urchin development and biomineralization. My project aimed to understand how the common actomyosin remodelling proteins such as Cdc42 and PAKs regulate biomineral growth and morphology in invertebrates.

Teaching Assistant, Department of Marine Biology**,** University of Haifa Israel. 2022- Pursuing

- I am currently serving as teaching assistant in two courses called 'Research Methods in Marine Biology A; and B'. These courses aim to familiarise students with important scientific questions and objectives and to design and conduct independent research projects.

ICAR-National Bureau of Plant Genetic Resources, Pusa Campus, New Delhi. Jan – Sept 2020
Bachelor's Thesis Project

Supervisor: Dr. Sandhya Gupta, Principal Scientist, Tissue Culture and Cryopreservation Unit
Thesis: Cryopreservation and assessment of genetic stability of cryopreserved *Fragaria* spp. using SSR markers.

- Independently executed interdisciplinary work in cryopreservation and molecular biology protocols on two different Fragaria species.

ICAR-National Bureau of Plant Genetic Resources, Pusa Campus, New Delhi. June-July, 2020
Summer Intern
Supervisor: Dr. Sandhya Gupta, Principal Scientist, Tissue Culture and Cryopreservation Unit
Title: Cryopreservation of *Fragaria sp*. using different techniques.
Performed different cryopreservation techniques on *Fragaria* x *annanasa.*

INTERSHIPS

Temperate Breeding and Research Station, **ACSEN HYVEG Pvt Ltd**, Kullu, India
Summer Intern June – July 2018
Supervisor: Dr. Arun Agarwal, Head, Biotechnology.
Studied and researched micropropagation on *Brassica oleracea* var. *capitata* (Cabbage) and various plant tissue culture techniques and protocols.

Mrd Life Sciences, Lucknow, India
Summer Intern June 2017
Supervisor: Dr. Amit Singh, Senior Scientist.
Hands on training on enzymology protocols and techniques.

PUBLICATIONS

- Chen, S., Gildor, T., ***Tewari, P***., & Ben-Tabou de-Leon, S. (2025). Skeletogenic Expression of Integrin Alpha, Talin and Npnt Genes and Npnt Role in Sea Urchin Skeletogenesis. Journal of Experimental Zoology Part B-Molecular and Developmental Evolution.
- Gupta, S. and ***Tewari, P***. (2020). Cryopreservation of *Fragaria* × *ananassa* using different techniques. Acta Horticulturae 1298, pp. 161-166.

LABORATORY SKILLS

- **Molecular Biology:** DNA and RNA extraction and estimation, Gel electrophoresis and quantification, Polymerase Chain Reaction and qPCR, RT-PCR, Gene cloning and transformation, cDNA preparation, Embryo culture, MASO and DNA Microinjections, preparation of competent cells *etc*.
- **Microscopy:** Fluorescence and Confocal microscopy (LSM 980), Spinning Disc Confocal, live imaging of marine embryos, etc.
- **Biochemistry:** Immunostaining, Whole mount *in-situ* hybridisation, *etc.*
- **Maintenance of aquaria system** for marine organism like sea urchins and microalgae.
- **Microbiology:** Different media preparation, isolation of microbes, obtaining pure culture through streaking, and antibiotic susceptibility test.
- **Cryopreservation**: Encapsulation-dehydration, vitrification, and cryoplates method.

ORAL and POSTER PRESENTATIONS

Indian Society of Developmental Biology-International Society of Differentiation joint meeting, IISER Berhampur, Odisha, India December 2025
Poster: *Uncovering Actomyosin Remodeling Mechanisms Regulating Mineral Trafficking in Biomineralization*

Eleventh Academic Get-Together of Department of Marine Biology, University of Haifa. June 2025
Poster: *Deciphering the cytoskeletal control of vesicular transport during sea urchin larval biomineralization*

28th Conference on Developmental Biology of Sea Urchins and Other Marine Invertebrates, Marine Biology Laboratory, Woods Hole, USA. April 2025
Oral: *p21 Activated Kinases Expression, Regulation, and their Roles in Sea Urchin Development and Skeletogenesis*
Israeli Society of Evolutionary Biology Annual conference, Hebrew University. October 2024
Poster: *p21 Activated Kinases Expression, Regulation, and their Roles in Sea Urchin Development and Skeletogenesis*
56th Annual Meeting of the Israeli Society of Microscopy, Ben Gurion University of Negev. May, 2024
Poster and Flash Talk: *p21 Activated Kinases Expression, Regulation, and their Roles in Sea Urchin Development and Skeletogenesis*
Tenth Academic Get-Together of Department of Marine Biology, University of Haifa. March 2024
Poster: *p21 Activated Kinases Expression, Regulation, and their Roles in Sea Urchin Development and Skeletogenesis*
Israeli Society of Evolutionary Biology Annual conference, University of Haifa. April 2023
Poster: *Actomyosin Remodelling Genes and their Role in Sea Urchin Larval Skeletogenesis*
EchinoClub Monthly Webinar Forums (Online). February 2023
Oral: *Role of p21 activated kinase (Pak) in Sea Urchin Larval Skeletogenesis*
Ninth Academic Get-Together of Department of Marine Biology, University of Haifa. January 2023
Oral: *Actomyosin Remodelling Genes and their Role in Sea Urchin Larval Skeletogenesis*
Eighth Academic Get-Together of Department of Marine Biology, University of Haifa. March 2022
Poster: *Spatial expression of cytoskeleton remodelling genes in the skeletogenic cells of the sea urchin embryo.*

AWARDS AND SCHOLARSHIPS

- **Best poster award** at the Eleventh Academic Get-Together of Department of Marine Biology, University of Haifa.
- **University of Haifa and School of Marine Sciences travel grant** to attend 28th Conference on Developmental Biology of Sea Urchins and Other Marine Invertebrates, Marine Biology Laboratory, Woods Hole, USA
- **Frist runner up at the 3-minute thesis competion** organised by Bloom School of Graduate Studies, University of Haifa.
- **Marom Scholarship for PhD** for academic year 2023-2027
- **Mairne School Exllency Scholarship for Phd** for the academic year 2023-2027

LEADERSHIP EXPERIENCE

Quiz master | Marine Biology Get together 2023 - till now
Organised a Fun activity session for fostering network and collaboration during the annual get together.
6th Cohort of Haifa Grad Team | Bloom School of Graduate Studies, UoH 2024-2025
Member
Organised Haifa Grad Team annual Conference on "Resilience in Academia" and a round table discussion on "Bridging gap in Academia and Industry".
Ninth Haifa Conference in Marine Sciences | Leon H Charney School of Marine Science, UoH June 2022
Core Committee member
Organised annual conference of marine conference, on Ocean and Climate Change, featuring 10 speakers and attracting 400+ participants from all over Israel. The conference also features talk from different NGOs working toward reducing climate change.
Society of Biotechnology | Jacob Institute of Biotechnology and Bioengineering, SHUATS. 2018 – 2019
President
Lead one of the largest clubs on campus with 7-person executive member and 250+ members. Organised a national conference on Recent Advances in Agriculture, Food Tech, and Human Health, featuring 25+ speakers and attracting 500+ attendees.

Executive member 2019- 2020

Planned and organised a two-days biotechnology skill development program for school students, featuring 10 panellists.

Entrepreneurs and Innovators Club | Directorate of Innovation, Project, and Consultancy, SHUATS. 2016-2019

Student Coordinator

Coordinated with 50 faculty members and 75 students to sensitise the campus about IP, entrepreneurship, and patent. Organised three World Intellectual Property Day, and 1 Entrepreneurship Awareness Camp.

Student Editor, Editorial Board 2018 – 2019

First student member on all faculty editorial board of quarterly newsletter of the club. Wrote and reviewed articles for two volumes of the journal on different themes like Innovations in Sports, Technology in Independent India, *etc.*

SKILLS AND INTEREST

- **Technical:** Proficient in MS Office (Word, PowerPoint, Excel) ImageJ, IMARIS, Abode Photoshop and Illustrator etc.
- **Languages:** Professional fluency in English and basic knowledge of German and French.
- **Interest:** Reading, swimming, oration, and painting.

PROFESSIONAL ASSOCIATIONS

- Indian Society of Developmental Biology 2025-Present
- European Society of Microscopy 2024-Present
- Israeli Society of Microscopy 2024-Present

REFERENCES

Prof. Smadar Ben Tabou de Leon
Head and Associate Professor,
Department of Marine Biology, Leon H. Charney School of Marine Sciences
University of Haifa. Haifa, Israel
Email: sben-tab@univ.haifa.ac.il

Dr. Tal Luzzatto Knaan,
Assistant Professor,
Department of Marine Biology, Leon H. Charney School of Marine Sciences
University of Haifa. Haifa, Israel.
Email: tluzzatto@univ.haifa.ac.il

Dr. Sandhya Gupta
Principal Scientist,
Division of Germplasm Conservation,
ICAR- National Bureau Plant Genetic Resources, Pusa Campus, New Delhi, India
Email: sandhya.gupta@icar.gov.in